

ARDUINO ARDUINO WEB EDITOR REVIEW ON DIGITAL I/O

University of Alexandria, Faculty of Engineering,
Computer and Systems Engineering Department.
CSE343 Embedded Systems Spring 2026

Arduino Web Editor

- The Arduino Web Editor allows you to write code and upload sketches to any official Arduino board directly from your web browser.
- This IDE (Integrated Development Environment) is part of Arduino Create, an online platform that enables developers to write code, access tutorials, configure boards, and share projects.
- Develop on cloud facility
 - Always be up-to-date with the latest features and support for new boards.
 - Automatically recognizes any Arduino and Genuino board connected to your PC, and configures itself accordingly.

Getting Started with Arduino Web Editor

1. Install the Arduino Create Agent plugin.
2. Create a new Arduino Account at this link. Complete the registration form, then hit the "create account" button. Then you will receive an email with a link to activate your account. Select the link and a new page will open with your confirmed account information.
3. Log in the Arduino Web Editor.

Arduino Web Editor Screen

The screenshot displays the Arduino Web Editor interface. The browser address bar shows the URL: `create.arduino.cc/editor/waheed1996/7ff2aa55-2f65-4b25-a856-42f553c12187`. The left sidebar contains navigation links: EDITOR, Sketchbook, Examples, Libraries, Monitor, Reference, Help, Preferences, and Features usage. The main editor area is titled "Blink" and shows the selected board as "Arduino Uno" and the port as "COM11". The code editor displays the "Blink.ino" file with the following content:

```
8 the correct LED pin independent of which board is used.
9 If you want to know what pin the on-board LED is connected to on your Arduino
10 model, check the Technical Specs of your board at:
11 https://www.arduino.cc/en/Main/Products
12
13 modified 8 May 2014
14 by Scott Fitzgerald
15 modified 2 Sep 2016
16 by Arturo Guadalupi
17 modified 8 Sep 2016
18 by Colby Newman
19
20 This example code is in the public domain.
21
22 https://www.arduino.cc/en/Tutorial/BuiltInExamples/Blink
23 */
24
25 // the setup function runs once when you press reset or power the board
26 void setup() {
27   // initialize digital pin LED_BUILTIN as an output.
28   pinMode(LED_BUILTIN, OUTPUT);
29 }
30
31 // the loop function runs over and over again forever
32 void loop() {
33   digitalWrite(LED_BUILTIN, HIGH); // turn the LED on (HIGH is the voltage level)
34   delay(1000); // wait for a second
```

Below the code editor, a green success message states: "Success: Saved on your online Sketchbook and done uploading Blink". At the bottom, the serial monitor shows the output: "avrdude: safemode: efuse reads as 0", "avrdude: safemode: Fuses OK (E:00, H:00, L:00)", and "avrdude done. Thank you." A "New Cloud Editor is out, try it now!" banner is visible in the bottom right corner.

Sketchbook

The screenshot displays the Arduino IDE Sketchbook interface. The top bar shows the URL `create.arduino.cc/editor/waheed1996/7ff2aa55-2f65-4b25-a856-42f553c12187`. The left sidebar contains navigation links: EDITOR, Sketchbook, Examples, Libraries, Monitor, Reference, Help, Preferences, and Features usage. The main editor area shows the 'Blink' sketch, which is selected in the top dropdown menu. The code is displayed in a text editor with line numbers 8 through 34. The code includes comments about the LED pin and the setup/loop functions. A green status bar at the bottom indicates 'Success: Saved on your online Sketchbook and done uploading Blink'. The bottom right corner features a 'New Cloud Editor is out, try it now!' notification.

create.arduino.cc/editor/waheed1996/7ff2aa55-2f65-4b25-a856-42f553c12187

UPGRADE PLAN

EDITOR

Sketchbook

Examples

Libraries

Monitor

Reference

Help

Preferences

Features usage

CLOUD

Blink

Arduino Uno COM11

Blink.ino

Blink.txt

```
8 the correct LED pin independent of which board is used.
9 If you want to know what pin the on-board LED is connected to on your Arduino
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28   pinMode(LED_BUILTIN, OUTPUT);
29 }
30
31 // the loop function runs over and over again forever
32 void loop() {
33   digitalWrite(LED_BUILTIN, HIGH); // turn the LED on (HIGH is the voltage level)
34   delay(1000); // wait for a second
```

Success: Saved on your online Sketchbook and done uploading Blink

avidude: safemode: efuse reads as 0
avidude: safemode: Fuses OK (E:00, H:00, L:00)
avidude done. Thank you.

New Cloud Editor is out, try it now!

Examples

The screenshot shows the Arduino IDE web editor interface. The left sidebar contains navigation links: EDITOR, Sketchbook, Examples (selected), Libraries, Monitor, Reference, Help, Preferences, and Features usage. The Examples section is expanded, showing a list of categories: 01.BASICS (6), 02.DIGITAL (9), 03.ANALOG (6), 04.COMMUNICATION (12), 05.CONTROL (6), 06.SENSORS (4), 07.DISPLAY (2), 08.STRINGS (14), 09.USB (7), 10.STARTERKIT_BASICKIT (14), and 11.ARDUINOISP (1). The main editor area displays the 'Blink' example code, which includes comments about the correct LED pin and the public domain status of the code. The code is as follows:

```
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9 If you want to know what pin the on-board LED is connected to on your Arduino
10 model, check the Technical Specs of your board at:
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32 void loop() {
33   digitalWrite(LED_BUILTIN, HIGH); // turn the LED on (HIGH is the voltage level)
34   delay(1000); // wait for a second
```

Below the code, a green banner displays the success message: "Success: Saved on your online Sketchbook and done uploading Blink". At the bottom, the serial monitor shows the output: "avidude: safemode: efuse reads as 0", "avidude: safemode: Fuses OK (E:00, H:00, L:00)", and "avidude done. Thank you.".

Libraries

The screenshot displays the Arduino IDE interface. On the left is a teal sidebar with navigation options: EDITOR, Sketchbook, Examples, Libraries (selected), Monitor, Reference, Help, Preferences, and Features usage. At the bottom of the sidebar are icons for a plus sign and the CLOUD logo.

The main workspace is divided into two panes. The left pane is the 'LIBRARY MANAGER' with a search bar and tabs for DEFAULT, FAVORITES, and CUSTOM. It shows libraries for the 'UNO' board, including 'ARDUINOBEARSSL', 'ARDUINOCLOUDTHING', and 'ARDUINODMX', each with an 'Examples' link.

The right pane is the code editor, titled 'Blink'. It shows the 'Blink.txt' file with the following code:

```
8 the correct LED pin independent of which board is used.
9 If you want to know what pin the on-board LED is connected to on your Arduino
10 model, check the Technical Specs of your board at:
11 https://www.arduino.cc/en/Main/Products
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34   delay(1000); // wait for a second
```

Below the code editor, a green status bar displays the message: 'Success: Saved on your online Sketchbook and done uploading Blink'. At the bottom of the IDE, a terminal window shows the upload progress: 'avrdude: safemode: efuse reads as 0', 'avrdude: safemode: Fuses OK (E:00, H:00, L:00)', and 'avrdude done. Thank you.'.

In the top right corner of the IDE, there is an 'UPGRADE PLAN' button, a cloud icon, and a user profile icon labeled 'AH'. A small notification bubble in the bottom right corner says 'New Cloud Editor is out, try it now!'.

Monitor

The screenshot shows the Arduino IDE interface with the Monitor window open. The left sidebar contains navigation links: EDITOR, Sketchbook, Examples, Libraries, Monitor (selected), Reference, Help, Preferences, and Features usage. The top toolbar shows the PORT dropdown set to 'Arduino Uno at COM11', the baud rate set to '9600 baud', and a 'SEND' button. The main editor area displays the 'Blink' sketch, which includes a header file 'Blink.h' and a main function 'void loop()' that toggles the built-in LED on and off every 1000ms. The Monitor window at the bottom shows the serial output, indicating a successful upload and execution of the Blink sketch.

create.arduino.cc/editor/waheed1996/7ff2aa55-2f65-4b25-a856-42f553c12187

UPGRADE PLAN

PORT: Arduino Uno at COM11

Newline 9600 baud SEND

Blink

Arduino Uno COM11

Blink.ino Blink.txt

```
8 the correct LED pin independent of which board is used.
9 If you want to know what pin the on-board LED is connected to on your Arduino
10 model, check the Technical Specs of your board at:
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33   digitalWrite(LED_BUILTIN, HIGH); // turn the LED on (HIGH is the voltage level)
34   delay(1000); // wait for a second
```

Success: Saved on your online Sketchbook and done uploading Blink

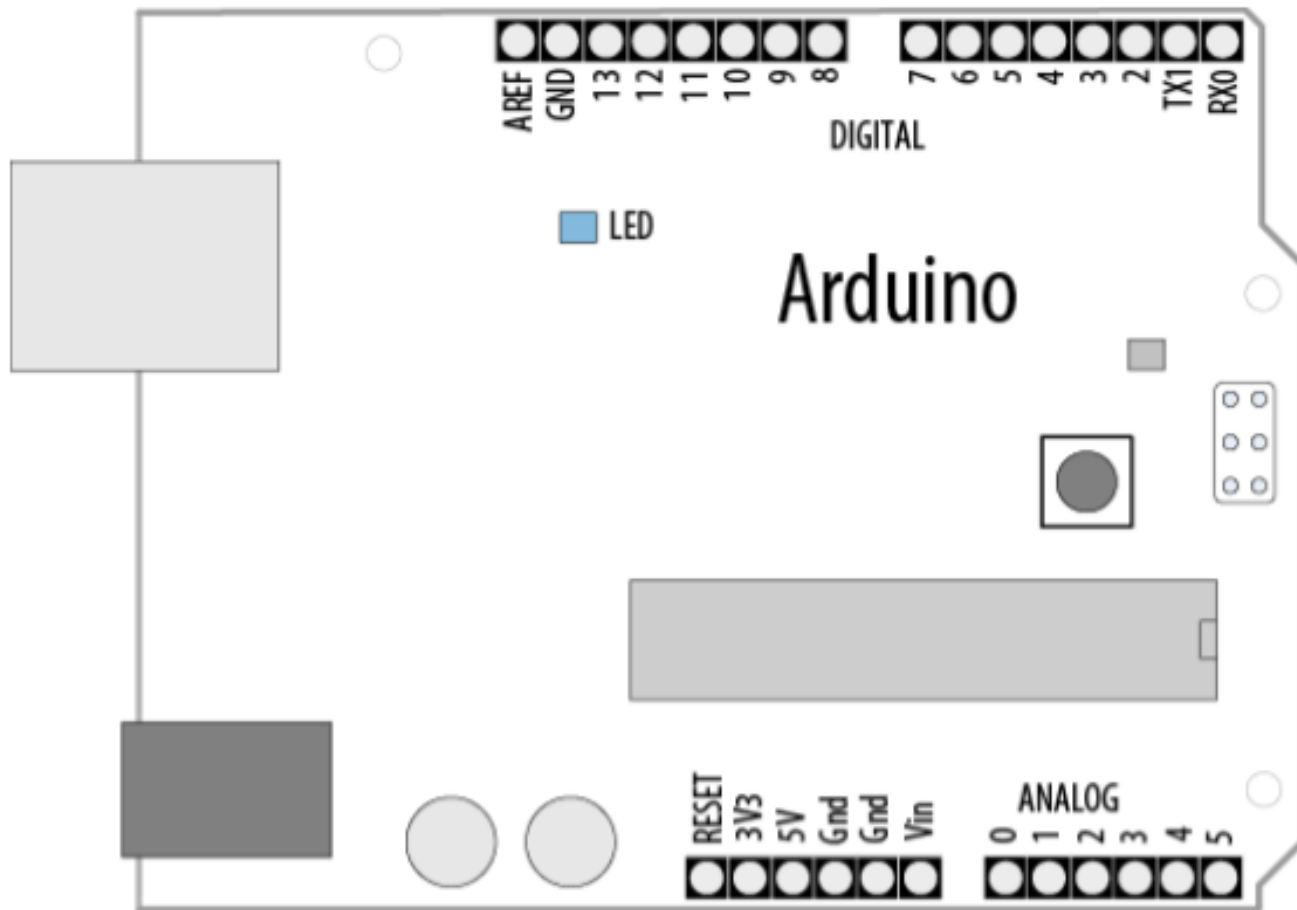
avridude: safemode: efuse reads as 0
avridude: safemode: Fuses OK (E:00, H:00, L:00)
avridude done. Thank you.

New Cloud Editor is out, try it now!

DIGITAL / ANALOG INPUT

- Arduino's ability to sense digital & analog inputs allows it to respond to you and the world around you
- Wiring the Arduino is simple and risk-free if you use the USB provided voltage (5 Volts).
- In case of higher voltages, the board may be damaged.
- Input & Output is done through the Analog and I/O pins

DIGITAL / ANALOG INPUT



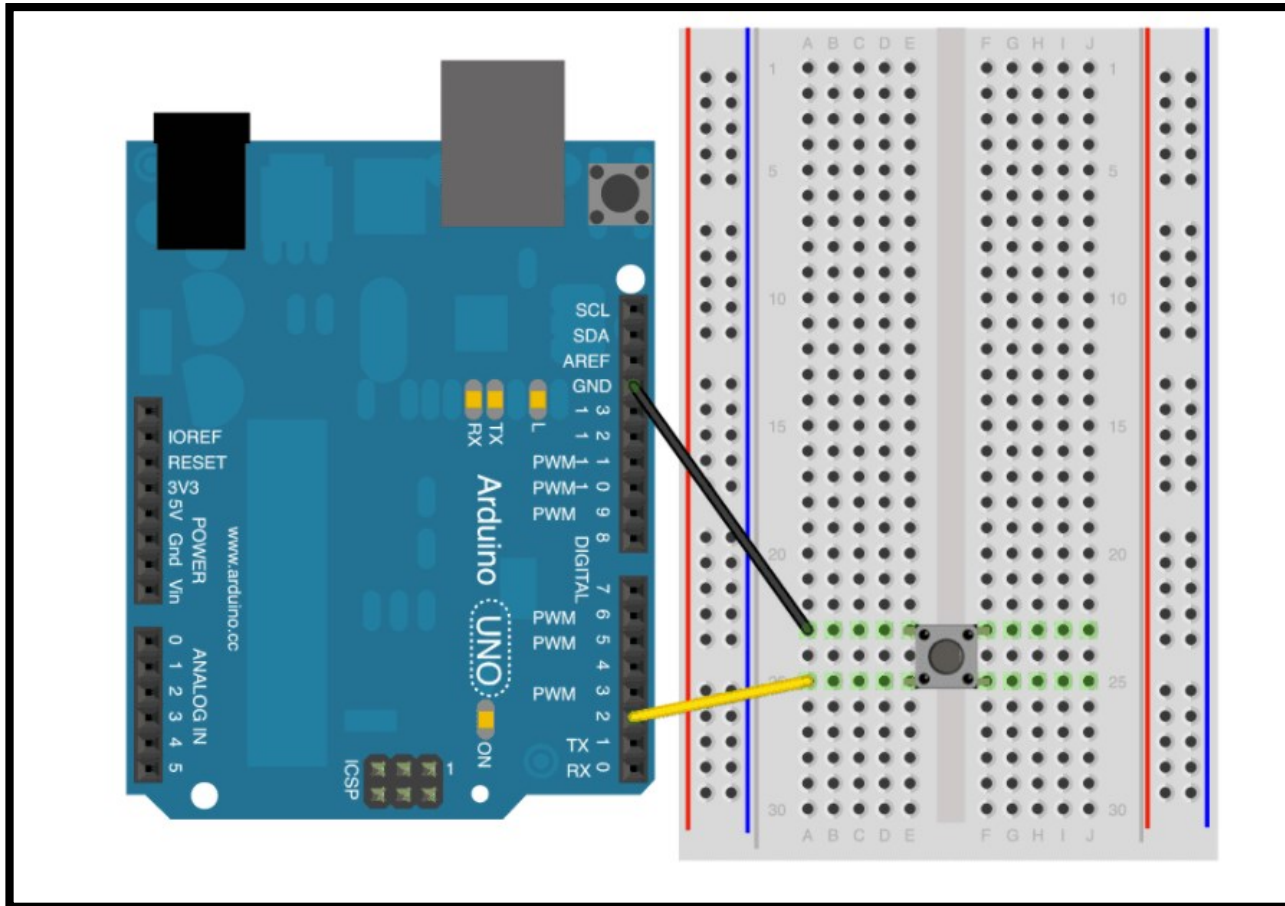
Arrangement of pins

Functions

- Some useful functions

Function	Description
<code>digitalRead()</code>	Read the voltage on a pin and state whether it is HIGH (5 Volts) or LOW (0 Volts).
<code>digitalWrite()</code>	Write the voltage to a pin and state whether it is HIGH (5 Volts) or LOW (0 Volts).
<code>pinMode()</code>	Used in setup to determine whether the pin will work as OUTPUT or INPUT

Interfacing a Push Button



Interfacing a Push Button(2)

```
void setup() {  
    //start serial connection  
    Serial.begin(9600);  
    //configure pin 2 as an input and enable the internal pull-up resistor  
    pinMode(2, INPUT_PULLUP);  
    pinMode(13, OUTPUT);  
}
```

Interfacing a Push Button(3)

```
void loop() {  
    //read the pushbutton value into a variable  
    int sensorVal = digitalRead(2);  
    //print out the value of the pushbutton  
    Serial.println(sensorVal);  
    // Keep in mind the pull-up means the pushbutton's Logic is inverted. It goes  
    // HIGH when it's open, and LOW when it's pressed. Turn on pin 13 when the  
    // button's pressed, and off when it's not:  
    if (sensorVal == HIGH) {  
        digitalWrite(13, LOW);  
    } else {  
        digitalWrite(13, HIGH);  
    }  
}
```

Challenges

- You want to avoid false readings due to *contact bounce* (contact bounce produces spurious signals at the moment the switch contacts close or open). The process of eliminating spurious readings is called *debouncing*.

Debouncing(1)

```
void loop() {  
    //read the pushbutton value into a variable  
    int sensorVal = digitalRead(2);  
    //print out the value of the pushbutton  
    Serial.println(sensorVal);  
    // Keep in mind the pull-up means the pushbutton's logic is inverted. It goes  
    // HIGH when it's open, and LOW when it's pressed. Turn on pin 13 when the  
    // button's pressed, and off when it's not:  
    if (debounce(2)) {  
        digitalWrite(13, LOW);  
    } else {  
        digitalWrite(13, HIGH);  
    }  
}
```

Debouncing(2)

```
boolean debounce(int pin) {  
    boolean state;  
    boolean previousState;  
    previousState = digitalRead(pin)  
    for (int i=0; i<10; i++) {  
        delay(1); // Wait for 1 millisecond  
        state = digitalRead(pin);  
        if (state != previousState) {  
            i = 0; // reset the iterative counter;  
            previousState = state; // set the previous state  
        }  
    }  
    return state;  
}
```

References

- **Arduino Web Editor Guide**

<https://docs.arduino.cc/learn/starting-guide/the-arduino-web-editor/>

- **Digital Input Pull-Up Resistor Demo**

<https://docs.arduino.cc/tutorials/generic/digital-input-pullup>